

Framing effects on consumer purchase intention: a mediation and moderation analysis

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Abstract

We test whether messaging frames (scarcity, sustainability, social proof, and a combined frame) increase purchase intention for everyday consumer products, whether this effect is mediated by perceived value, and whether it is moderated by price sensitivity. Data, code, and outputs all live in the same git repository as this manuscript.

1 Introduction

Mediation and moderation logic follows the classical framework [Baron and Kenny, 1986] and the regression-based approach in Hayes [2017].

2 Methods

We analysed a simulated consumer decision-making dataset of 300 participants, each randomly assigned to one of 5 message-framing conditions (a plain control plus scarcity, sustainability, social-proof, and combined frames). After viewing the framed product, participants reported multi-item measures of price sensitivity, sustainability concern, brand trust, and perceived value, followed by their purchase intention and a hypothetical buy decision.

Multi-item scales were averaged into composite scores after reverse-scoring the relevant brand-trust item, and a value of -99 was treated as missing. Data were cleaned with scripted, version-controlled steps; the full codebook and cleaning code live in the project repository. All models were estimated in R with ordinary least squares (`lm`). Every figure and table in this paper is generated directly from the analysis code, so the reported numbers cannot drift from the analysis.

3 Results

3.1 Main effect

Relative to the control frame, the combined frame raised purchase intention by 0.58 points on the seven-point scale ($p = 0.008$). Figure~1 shows the full pattern across conditions.

3.2 Mediation through perceived value

The frames also shifted perceived value upward, and perceived value strongly predicted purchase intention (Figure~2); adding it to the outcome model attenuated the condition coefficients, the signature of an indirect path.

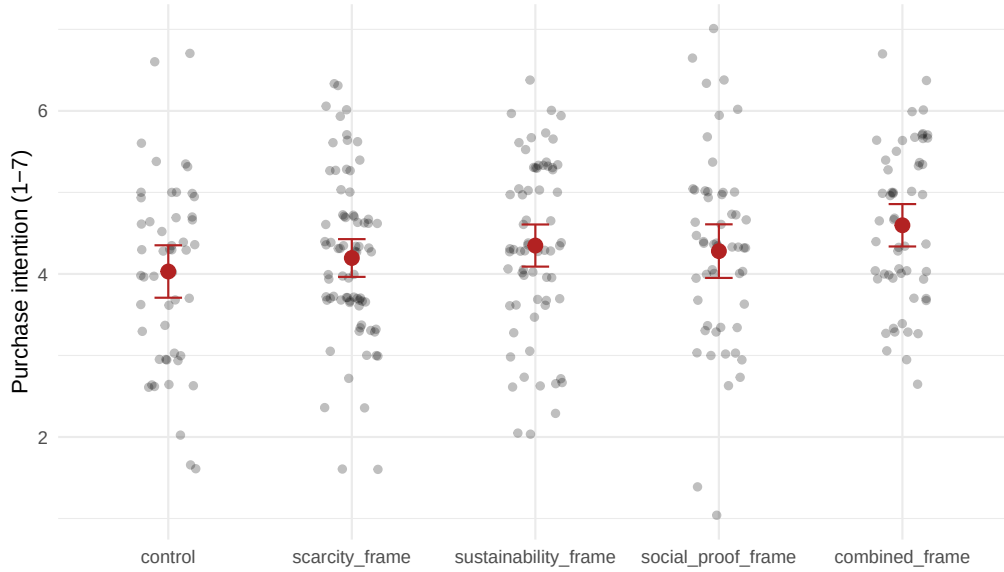


Figure 1: Purchase intention by condition. Red points are condition means with 95% confidence intervals.

3.3 Moderation by price sensitivity

The combined-frame boost was attenuated among the most price-sensitive respondents (Figure 3).

3.4 Model estimates

Table 1 collects the coefficients behind the figures above. It is generated by `reports/manuscript/results.qmd` and written into `generated/tabs/`, so the paper’s numbers track the analysis exactly.

4 Discussion

The figures and the model table above are produced by `reports/manuscript/results.qmd`, which reuses the same R functions as the project website. Re-rendering the manuscript recipes overwrites the files in `generated/`; this paper picks up the new versions on the next compile. No result is duplicated between the analysis code and the paper.

References

- Reuben M. Baron and David A. Kenny. The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6):1173–1182, 1986. doi: 10.1037/0022-3514.51.6.1173.
- Andrew F. Hayes. *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford Press, New York, NY, 2nd edition, 2017.

Table 1: Model estimates with 95% confidence intervals.

Model	Term	Estimate	SE	CI low	CI high	p
total	(Intercept)	4.08	0.25	3.59	4.58	0.000
total	conditionscarcity_frame	0.14	0.20	-0.26	0.54	0.494
total	conditionsustainability_frame	0.31	0.21	-0.09	0.72	0.133
total	conditionsocial_proof_frame	0.24	0.22	-0.19	0.67	0.271
total	conditioncombined_frame	0.58	0.22	0.15	1.01	0.008
total	age	0.00	0.01	-0.01	0.01	0.858
total	genderman	-0.03	0.14	-0.30	0.24	0.816
total	gendernon_binary	-0.46	0.34	-1.14	0.21	0.175
total	genderprefer_not_to_say	0.12	0.30	-0.46	0.71	0.676
total	income_group.L	-0.18	0.13	-0.43	0.08	0.177
total	income_group.Q	0.02	0.10	-0.18	0.23	0.841
mediator	(Intercept)	4.06	0.24	3.58	4.55	0.000
mediator	conditionscarcity_frame	0.04	0.20	-0.34	0.43	0.820
mediator	conditionsustainability_frame	0.48	0.20	0.09	0.87	0.016
mediator	conditionsocial_proof_frame	0.20	0.21	-0.22	0.61	0.351
mediator	conditioncombined_frame	0.85	0.21	0.44	1.26	0.000
mediator	age	0.00	0.01	-0.01	0.01	0.877
mediator	genderman	-0.11	0.13	-0.37	0.15	0.413
mediator	gendernon_binary	-0.07	0.33	-0.72	0.58	0.832
mediator	genderprefer_not_to_say	0.43	0.29	-0.13	1.00	0.132
mediator	income_group.L	-0.10	0.12	-0.35	0.15	0.424
mediator	income_group.Q	0.04	0.10	-0.15	0.24	0.665
outcome_w_m	(Intercept)	2.23	0.32	1.60	2.87	0.000
outcome_w_m	conditionscarcity_frame	0.12	0.18	-0.24	0.48	0.516
outcome_w_m	conditionsustainability_frame	0.09	0.19	-0.28	0.46	0.630
outcome_w_m	conditionsocial_proof_frame	0.15	0.20	-0.24	0.54	0.443
outcome_w_m	conditioncombined_frame	0.19	0.20	-0.20	0.59	0.334
outcome_w_m	perceived_value	0.46	0.06	0.35	0.57	0.000
outcome_w_m	age	0.00	0.01	-0.01	0.01	0.784
outcome_w_m	genderman	0.02	0.12	-0.22	0.26	0.888
outcome_w_m	gendernon_binary	-0.43	0.31	-1.04	0.17	0.160
outcome_w_m	genderprefer_not_to_say	-0.07	0.27	-0.60	0.46	0.786
outcome_w_m	income_group.L	-0.13	0.12	-0.36	0.10	0.267
outcome_w_m	income_group.Q	0.00	0.09	-0.18	0.19	0.990
moderation	(Intercept)	4.15	0.24	3.67	4.63	0.000
moderation	conditionscarcity_frame	0.09	0.20	-0.30	0.48	0.649
moderation	conditionsustainability_frame	0.27	0.20	-0.12	0.66	0.177
moderation	conditionsocial_proof_frame	0.09	0.21	-0.33	0.51	0.667
moderation	conditioncombined_frame	0.48	0.21	0.06	0.89	0.025
moderation	price_sensitivity_c	-0.49	0.12	-0.72	-0.25	0.000
moderation	age	0.00	0.01	-0.01	0.01	0.947
moderation	genderman	-0.07	0.13	-0.33	0.19	0.603
moderation	gendernon_binary	-0.32	0.33	-0.97	0.33	0.329
moderation	genderprefer_not_to_say	0.15	0.29	-0.41	0.72	0.590
moderation	income_group.L	-0.21	0.13	-0.45	0.04	0.102
moderation	income_group.Q	0.03	0.10	-0.16	0.23	0.743
moderation	conditionscarcity_frame:price_sensitivity_c	0.29	0.15	-0.01	0.60	0.060
moderation	conditionsustainability_frame:price_sensitivity_c	0.28	0.16	-0.05	0.60	0.095
moderation	conditionsocial_proof_frame:price_sensitivity_c	0.24	0.16	-0.08	0.57	0.142

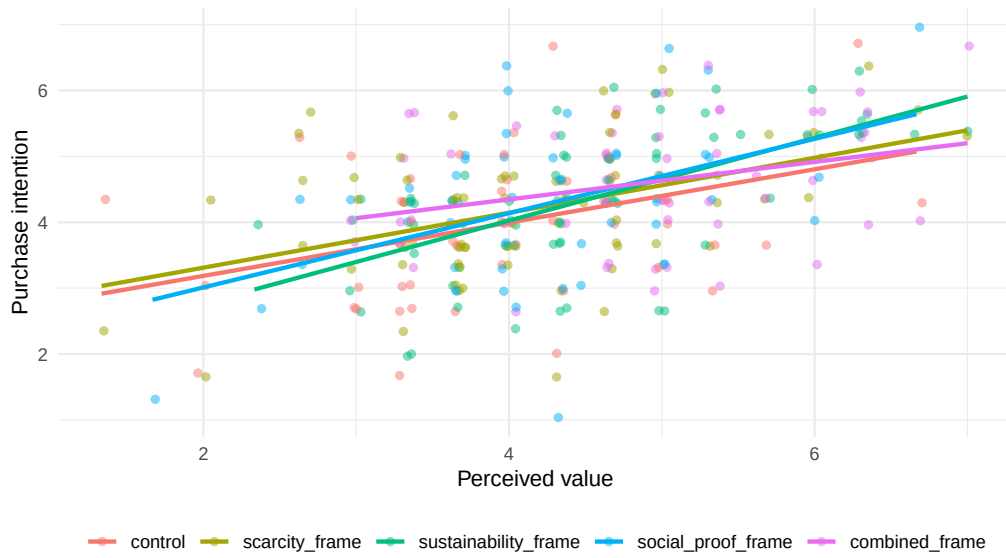


Figure 2: Perceived value predicts purchase intention; the frames shift perceived value upward, producing an indirect path.

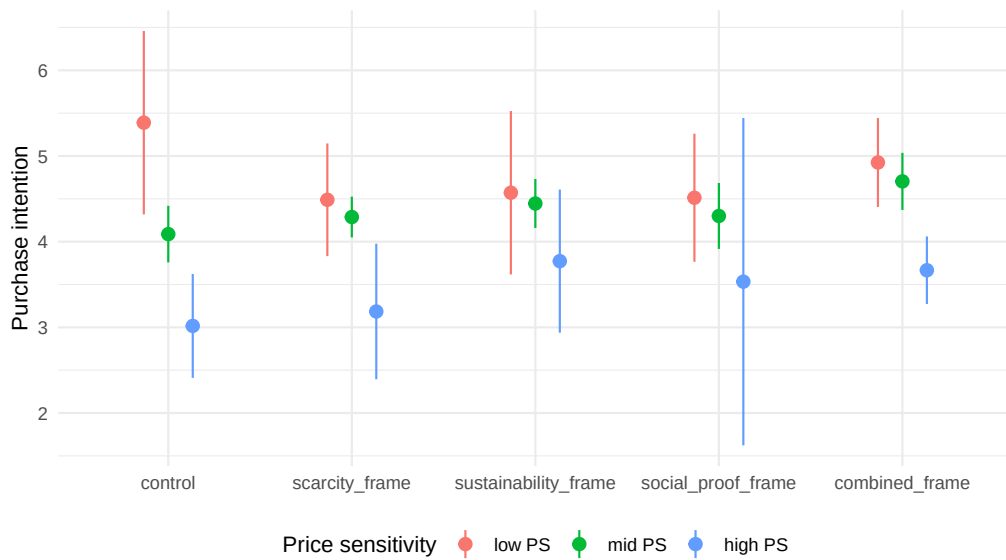


Figure 3: Condition $\times$ price sensitivity: the combined-frame boost shrinks for high-price-sensitive respondents.